

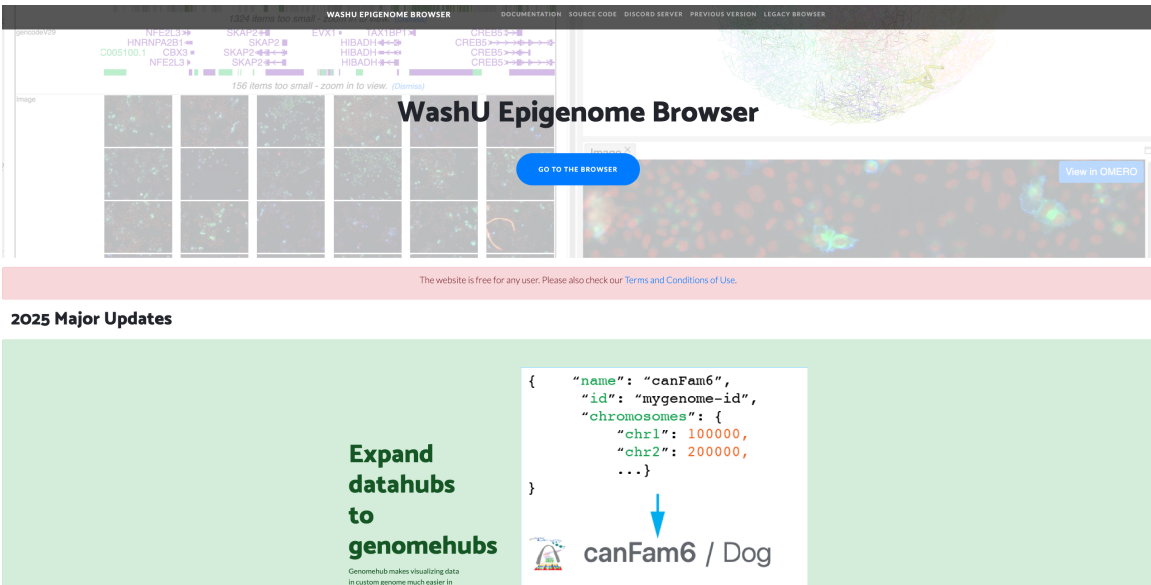
Tutorial for visualizing pedigree T2T datasets in the WashU Epigenome Browser

The haplotype-resolved genomes generated in this study have been incorporated into the Pedigree T2T collection in the WashU Epigenome Browser, providing an interactive platform for visualization of complete telomere-to-telomere assemblies together with centromere annotations and long-read epigenomic datasets. Each assembly includes reference sequence, gene annotation, RepeatMasker annotation, and cenSat annotation. Associated epigenomic datasets can be loaded through the public data hub for comparative visualization across cell types.

Step 1. Open the WashU Epigenome Browser

Navigate to: <https://epigenomegateway.wustl.edu/>

Click “Go to Browser” to enter the genome browser.

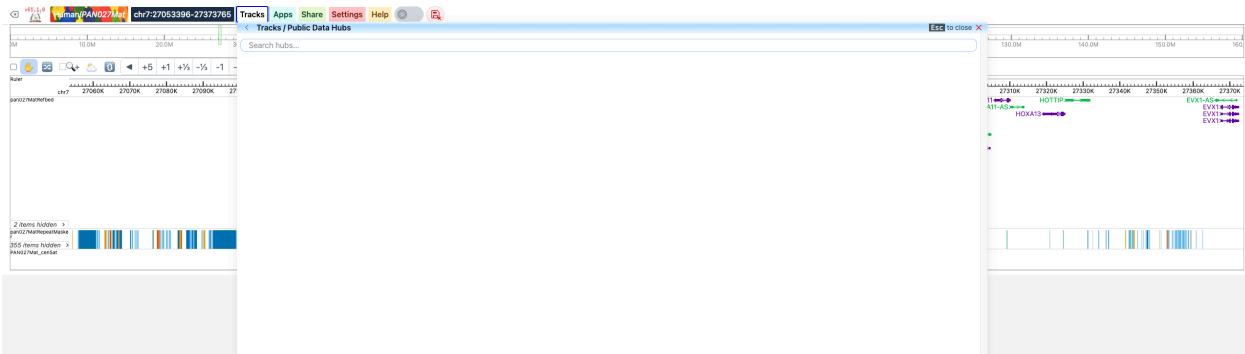


Step 2. Select a pedigree T2T assembly

From the **Genome** menu, select **Pedigree T2T**, then choose any of the eight haplotype-resolved assemblies generated in this study. Here we use **PAN027Mat** as an example.

- iPSC
- NPC

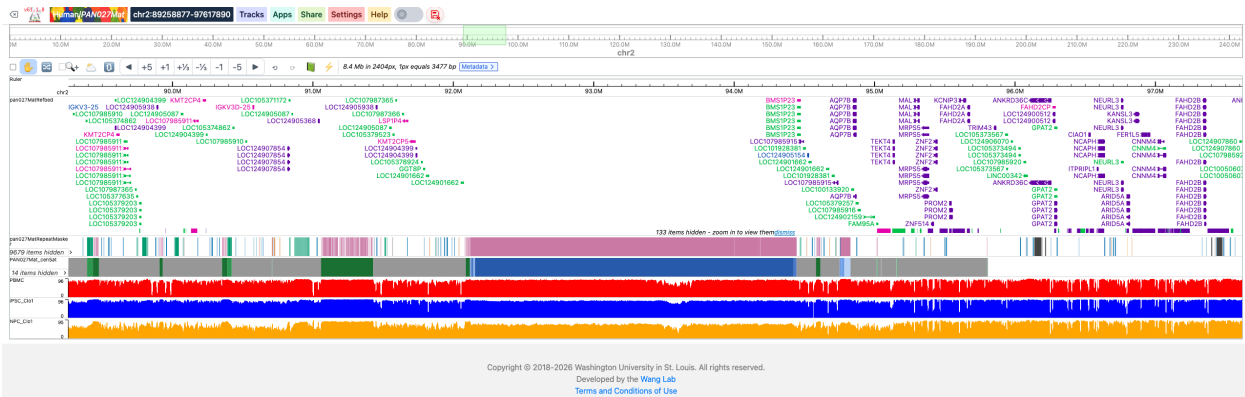
for the corresponding assembly.



Step 5. Explore centromere methylation at multiple scales

Navigate to the centromeric region of interest (for example, chromosome 2).

At the megabase scale, methylation profiles can be compared across multiple cell types to visualize large-scale remodeling within active HOR arrays.



Zooming further into individual loci enables visualization of CpG methylation at near single-CpG resolution, allowing direct inspection of methylation patterns across individual genomic regions.

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48 **Notes**

49 The browser resource will continue to be updated as additional datasets become available.

50 Future releases will incorporate additional long-read epigenomic datasets, including Fiber-seq

51 and CENP-A DiMeLo-seq tracks, together with other centromere-associated functional

52 genomics data generated from the Pedigree T2T collection.

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